

THE COMPANY

Aptiv is an automotive technology supplier, a spin off of General Motors. Aptiv designs, develops and manufactures software and hardware solutions to enable a safer, greener and more connected future of mobility. Their clientele involves automobile OEMs around the world, helping them create vehicles with advanced safety features, electrified architectures and intelligent connectivity.

Aptiv provides solutions that take into account the vehicle's entire architecture – its software, hardware and electrical/electronic architecture. Aptiv's various business units work together to provide turnkey solutions for their customers.

Aptiv's most prominent business units (domains of work) include **Advanced Safety and User Experience (AS&UX)** and **Signals and Power Systems (S&PS)**. AS&UX deals with ADAS systems, building hardware agnostic platforms, interior systems and developing OTA software. S&PS delivers the high-speed data and high-power electrical systems required for feature-rich, highly-automated vehicles.

THE ROLE

Software Engineering Intern (Associate Embedded Software Engineer)

Responsibilities (as taken from the company's JD):

- Follow existing AS&UX standards in SW development process
- Collecting and analyzing SW requirements
- Preparing SW design documentation
- Programming SW modules
- Preparing and performing tests to verify SW work products.
- Continuous development of professional qualifications.
- Care for a good company image.
- Care for confidential information and company's property.

THE PROCESS

1. **Online Assessment (In Offline Mode)**
2. **Technical Interview**
3. **Techno-Managerial Interview**
4. **HR Interview**

THE TEST

The online test was taken by close to 200 people (CS,IT,ECE,EEE,El). It was conducted on HackerRank and the students were to report to CUIC to attend it. The test consisted of 18 questions:

1. Snippet / Mini Coding (finding output given a code snippet) – 2(10 mins)
2. MCQs (IT) – 9 (20 mins)
3. MCQs (Core) – 5 (10 mins)
4. Programming Question (Medium) – 1 (20 mins)
5. Programming Question (Advanced) – 1 (30 mins)

Programming Questions :

1. There are n points on the x -axis $(1, 2, 3, \dots, n)$, where the i th point has a cost denoted by $\text{cost}[i]$. Starting from coordinate $x = 0$, jumps of length k can be taken. Wherever a stop is made, the cost of that point is incurred. Find the minimum cost to reach the point n , taking jumps of length at most k .

2. You are organizing an event where there will be a number of presenters. The event starts at time 0 and time will be allocated for networking at any time during the event when there is not a presentation being made. The presentations may not overlap but can run consecutively, without breaks. While the order of speeches cannot be changed, there is a maximum number given that indicates how many speeches may be rescheduled. Maximize the length of the longest networking period.

MCQs : Operating Systems, Memory Management in Programming Languages, OOPs

Around 90 students were shortlisted to attend the interviews in-person.

THE INTERVIEW

The interviews were held in-person at CUIC. The Pre-Placement Talk (PPT) preceded the interviews and owing to quite a large number of students, the interview process went on for two days. I attended the first round on the same day as the PPT and the next round on the next day.

TECHNICAL ROUND (90 mins - 120 mins):

The first round was easily the longest interview that I had attended so far. The interviewer introduced himself and asked me to do the same. After a quick introduction, he jumped into the questions. Differences between a microprocessor & microcontroller, Memory management in C/C++ vs Java vs Python, How addressing works in n -bit architectures, Deadlocks, Process Scheduling, Interrupt servicing, Context switching and some basic bit manipulation questions were put forth to me.

Questions based on my resume were intertwined in between these. The interviewer was quite interested about my NLP project and asked me to explain about it like how I would to a person with no background in AI/ML. Questions were posed regarding my choice of going with a Neural Network compared to traditional ML algorithms. The interviewer was quite interested to know how one could use Neural networks in the automotive industry. He then asked me to write pseudocodes for some simple array-based and OOPs questions.

We finished the interview after he answered some questions I had for him. He asked for my location preferences as well.

TECHNO-MANAGERIAL ROUND (60 mins - 75 mins):

The second round took place the next day for me. After the introductions were exchanged, the interviewer questioned me about Operating Systems. Every concept was explored and I was asked a real-life example where each of the concepts could be applied. Questions about my knowledge of the company were asked. The interviewer wanted to know my understanding of how the concepts we've studied and practiced in college would be implemented in the automotive industry. He then asked for my GPA and after answering the questions I had for him, the interview was over.

HR ROUND (30 mins - 45 mins):

The HR round was the last round in this interview process. No technical questions were posed. Every question was aimed to illustrate the type of person I am, how I would react in certain scenarios, and my life outside of academics. The interview ended at around 7:00 PM in the evening and the results were to be announced in the upcoming days.

THE EXPERIENCE

The interview being held over the span of two days is something that could pose as a distraction and potentially mess you up whilst waiting in between rounds. But these things are quite the norm. You might get pushed up ahead or made to wait even longer. Everything will not go according to the schedule. But that is immaterial to the selection process. I was the last person to be interviewed by the panels in both the 2nd and 3rd rounds. So you might end up waiting almost an entire day for your chance, or you could be fast tracked till the last round with little to no gaps in between the first couple of rounds. Be ready for both.

Make sure to articulate your thoughts properly. There is no harm in taking a pause and having a sip of water while the interview is going on. Explain every aspect of your projects. Be it a minor bug you faced or making an informed decision regarding the choice of a model/algo. Have an answer ready for "Why Aptiv?" or "Why XYZ" in general. This was posed to me by all three of the interviewers.

Certain topics of importance when it comes to Aptiv are:

- Memory Management (stack/heap/garbage collection)
- Operating Systems (scheduling, deadlocks)
- Bitwise operators & Bit manipulation
- String & Array manipulation
- Microprocessors & Microcontrollers (8051 will make do)
- Data Structures (in the context of OS – memory allocation, file systems)

Be thorough with everything that you mention in your resume. You are expected to explain a lot when you give your answer to the questions being asked. The interviews are more of a conversation than a question-answering session.

My experience was quite a positive one. All of the interviewers were interested to listen to my responses. The first interviewer even gave me a couple of chocolates at the end lol. Even though the 2nd and 3rd rounds clashed with another exam I was due to attend, the panel members were understanding and waited till I finished the test.

The results were announced a couple of days later and I was one of the 15 students who were selected.

Be confident in your answers guys! Face the panel with a positive outlook on everything.